Template Week 5 – Operating Systems

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Assignment 5.1: Unix-like

a) Find out what the difference is between UNIX and unix-like operating systems?
   - UNIX is the original operating system created at Bell Labs. Unix-like operating systems were developed later but behave in a similar way and follow the same design principles. They are not the original UNIX but function similarly. Examples of Unix-like systems include Linux, macOS, iOS, and Android.

b) Study the image above named UNIX timeline. Find out who Ken Thompson, Dennis Ritchie, Bill Joy, Richard Stallman, and Linus Torvalds are and what they have contributed to the development of UNIX or unix-like systems and to IT in general. **TIP!** English-language sources often contain more detailed information about these individuals.
   - **Ken Thompson** co-created the original UNIX operating system at Bell Labs. He designed many of its early tools and helped establish the basic structure and ideas current Unix-like systems still use.
   - **Dennis Ritchie** worked with Thompson on UNIX and created the C programming language. Rewriting UNIX in C made it portable and influential. His work shaped almost all modern operating systems.
   - **Bill Joy** developed important parts of BSD UNIX, including early networking features and the vi editor. He later co-founded Sun Microsystems, helping bring UNIX-based systems into wider use.
   - **Richard Stallman** founded the GNU Project and created free, open-source replacements for many UNIX tools. GNU software is used together with the Linux kernel in most Linux distributions today.
   - **Linus Torvalds** created the Linux kernel in 1991. Linux is a widely used Unix-like operating system that runs on servers, phones, and many other devices. He also created Git, a major software development tool.

   - Together, these people shaped the evolution of UNIX and Unix-like systems. Their work forms the foundation of modern operating systems such as Linux, BSD, macOS, and Android.

c) What is the philosophy of the GNU movement?
   - The philosophy of the GNU movement focuses on software freedom for users. The GNU Project states that users should be free to use programs for any purpose, to study how they work, to change them, and to share them with others. These freedoms are meant to keep software open, ethical, and user controlled.
   - The GNU Project began in the 1980s with the goal of creating a free Unix-like system. According to the official GNU philosophy page, the deeper purpose is to protect user rights and encourage sharing, learning, and collaboration. Because of this philosophy, today's Linux systems use many tools from the GNU Project to work properly.

d) Does Ubuntu as a Linux operating system conform to the philosophy of the GNU movement? Please explain your answer.

- Ubuntu aligns with GNU's principles by using Linux and many GNU tools, and most of its software is open source. However, Ubuntu doesn't fully follow the strict GNU philosophy because it includes some closed-source drivers and firmware.

e) Find out what is the Windows Subsystem for Linux?
   - The Windows Subsystem for Linux (WSL) is a feature in Windows that allows you to run a Linux environment directly on a Windows computer. It lets you use Linux commands, tools, and even full Linux distributions (like Ubuntu) without needing a virtual machine or separate Linux installation.

f) Find out, which operating system family belongs to Android, iOS and ChromeOS?
   - Android, iOS, and ChromeOS are all Unix-like operating systems because they are built on Linux or UNIX-based foundations.

Assignment 5.2: Supercomputers and game consoles.

a) Research on this site what supercomputers are used for and write a short summary of it: https://www.computerhistory.org/timeline/search/?q=Supercomputer
   - Supercomputers are extremely powerful computers used to solve problems that are too complex for regular computers. They are commonly used for scientific research, weather forecasting, climate modeling, space exploration, and advanced simulations. Supercomputers can process very large amounts of data and perform calculations at very high speeds.

b) IBM is a company that has already built a number of supercomputers. One of them is IBM's Roadrunner. The CPU developed for this supercomputer was further developed at a later stage as the CPU for the PlayStation 3 console. Find out what a **PlayStation 3 cluster** is and what it was used for?
   - A PlayStation 3 cluster is a group of PlayStation 3 consoles connected to work as a supercomputer. Because the PS3 used the Cell processor, it was possible to use it for high-performance computing tasks. These clusters were used for scientific research such as protein folding, physics simulations, and military research.

c) You can build a supercomputer by putting a few computers together in a cluster. Here's what Oracle did with a collection of Raspberry Pi's, for example: https://blogs.oracle.com/developers/post/building-the-worlds-largest-raspberry-pi-cluster
   What specific operating system is running on this cluster?
   - Oracle's Raspberry Pi cluster runs a Linux-based operating system. Linux is commonly used in clusters because it is stable, flexible, and well suited for distributed computing.

d) Does Oracle's Raspberry Pi supercomputer appear in the list of the 500 fastest supercomputers in the world? Make a logical decision for this, without going through the entire list. https://www.top500.org/lists/top500/list/2023/06/

- No, Oracle's Raspberry Pi cluster does not appear in the Top500 list. The Top500 focuses on the fastest supercomputers in the world, while Raspberry Pi clusters are mainly built for education, experimentation, and demonstration rather than maximum performance.

e) What CPU architecture is used for the PlayStation 5 and Xbox Series X?
   - Both the PS5 and Xbox Series X use x86-64 CPU architecture based on AMD processors.

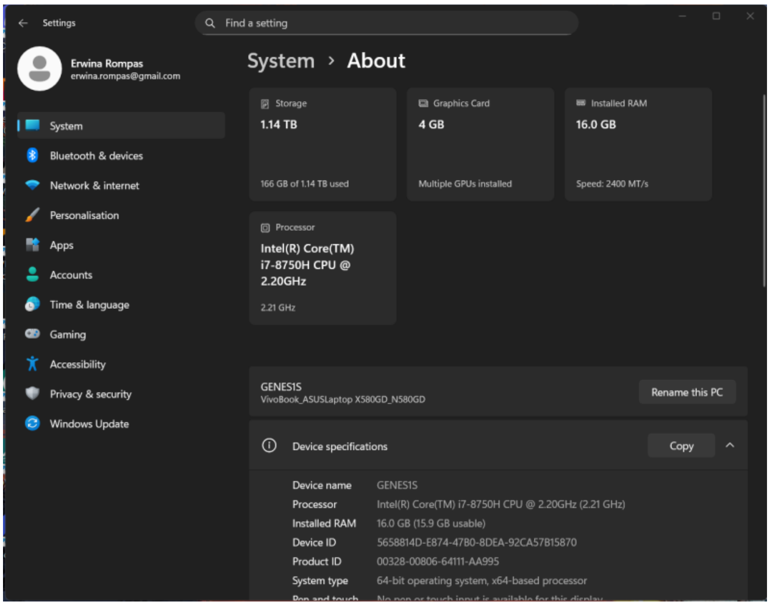
   What operating systems run on these consoles?
   - Both consoles run custom operating systems based on modified versions of existing systems. The PS5 uses a custom OS derived from FreeBSD, while the Xbox Series X uses a customized version of Windows.

   What conclusion can you draw from the answer to the previous question?
   - Modern game consoles use hardware and architecture similar to regular PC, which allows them to deliver high performance while remaining flexible for developers.

Assignment 5.3: Working with Windows

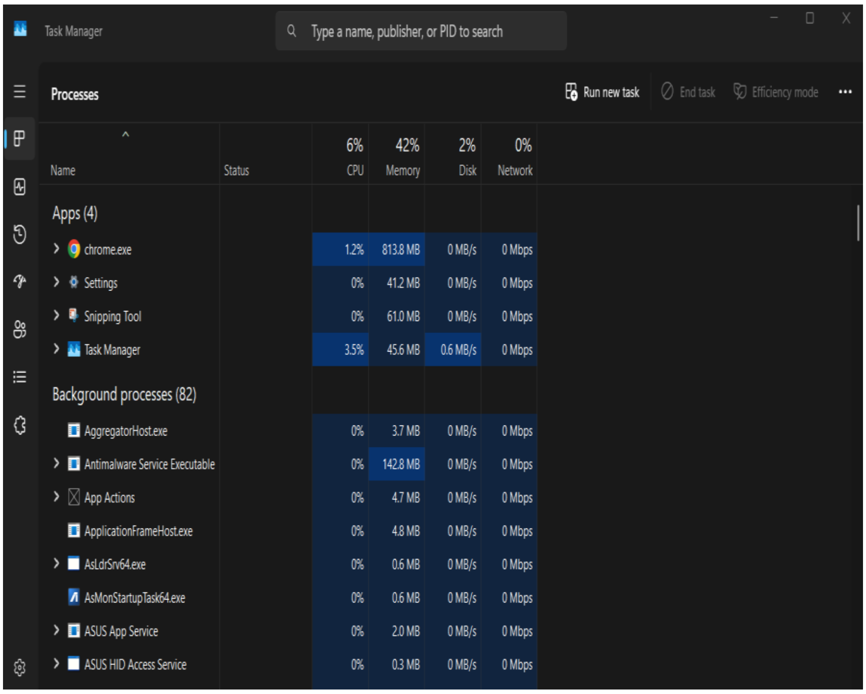
Take relevant screenshots of the assignments below

a) Practice for about 10 minutes with the keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.
b) The file explorer can be opened with + E, Which key combination could you also use?
   - File Explorer can also be opened with Win + X, then go to down, and click the File Explorer in the taskbar.
c) Open the system properties with a key combination, take a screenshot of the open screen. Paste this screenshot into this template.

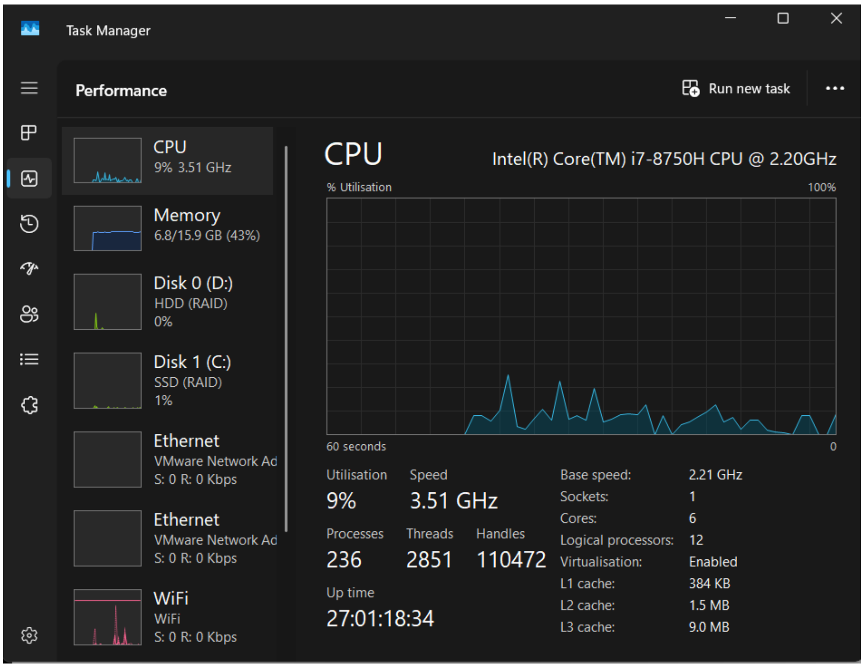


d) Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.

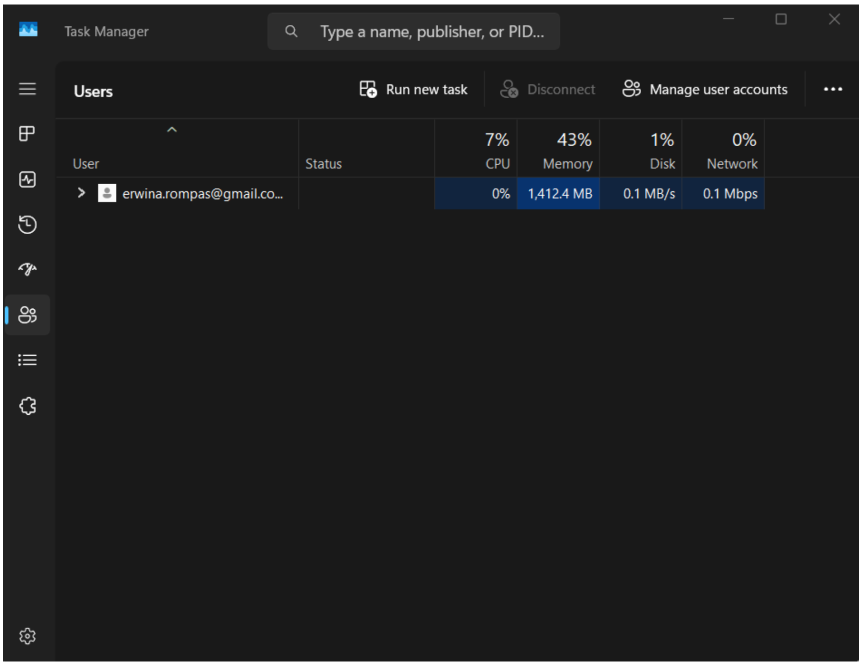
Process Tab



Performance Tab



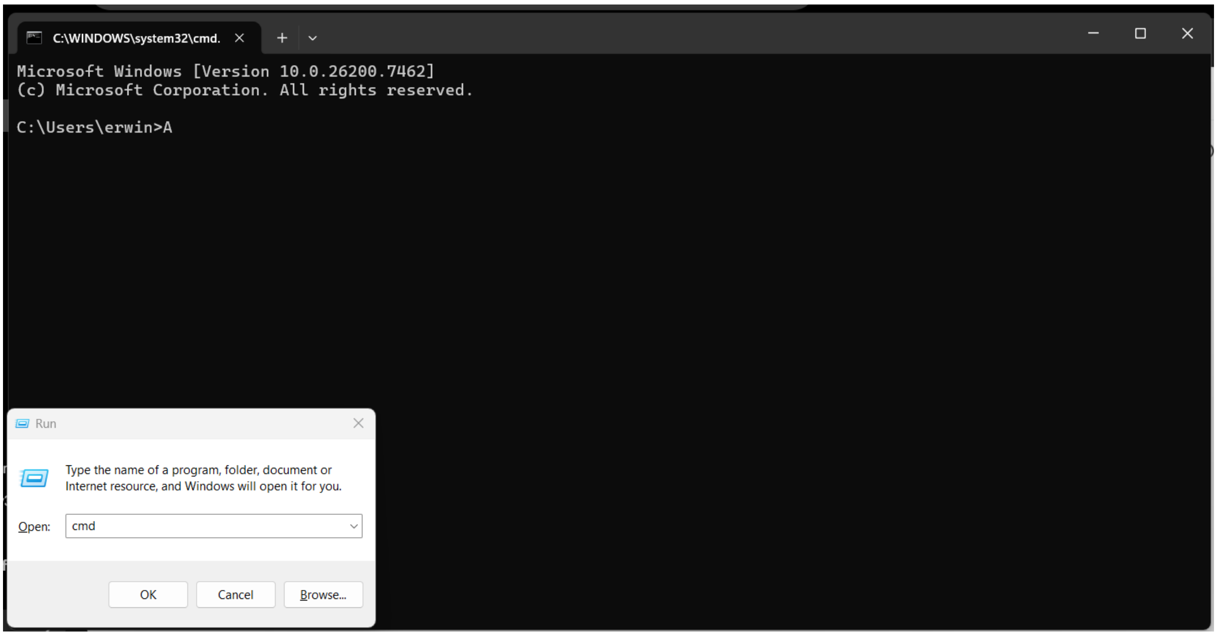
Users Tab



e) If you're giving a PowerPoint presentation and you connect your laptop to a projector, Windows can use the projector as a second screen. For example, you may have Outlook open on your first screen that you don't show over the projector, while the PowerPoint presentation is displayed on the projector, or the second screen. Which key combination should you use for this?
   - To use a projector or second screen, the key combination Win + P us used.

f) If you leave the classroom for a while and you leave your laptop behind, it is wise to lock the screen. Your Apps will continue to run in the background. So, for example, if you're waiting for a download that takes a while, lock the screen and get a cup of coffee. Which key combination do you use for this?
   - To lock the screen, use the combination Win + L.

g) Open the Run screen with a key combination. On this screen, type CMD and press <enter>. Take a screenshot of this result and paste it into this template.



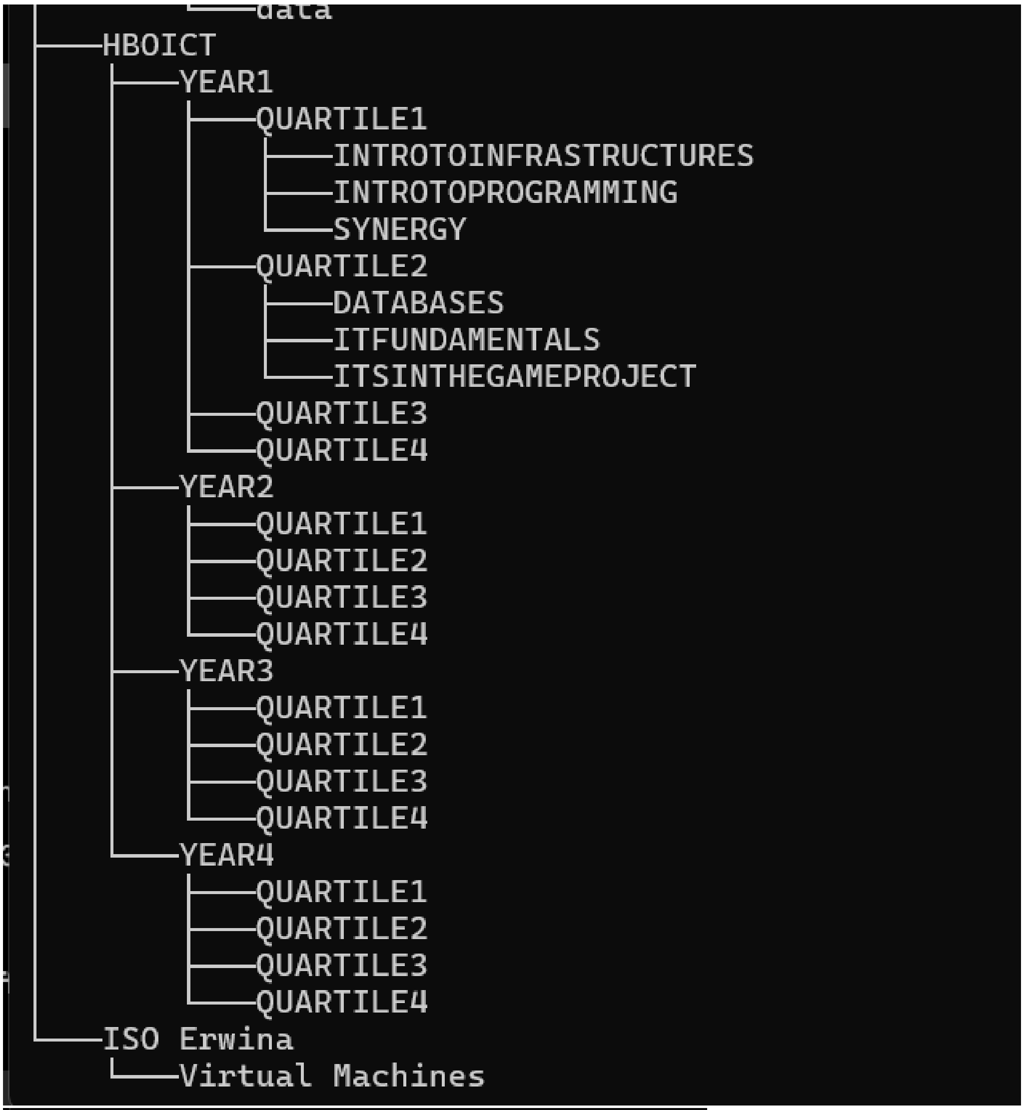
Working in the File Explorer

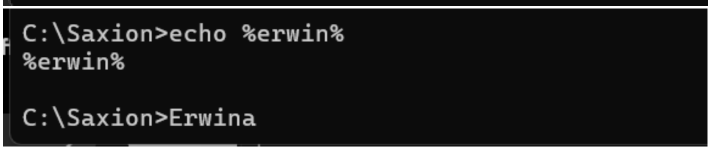
Relevant screenshots **copy** command:

```
C:\Saxion>copy Wave.png C:\Saxion\HBOICT\YEAR1\QUARTILE1\INTROTOPROGRAMMING
        1 file(s) copied.

C:\Saxion>copy Plug.png C:\Saxion\HBOICT\YEAR1\QUARTILE1\INTROTOINFRASTRUCTURE
        1 file(s) copied.

C:\Saxion>copy Tumble.png C:\Saxion\HBOICT\YEAR1\QUARTILE1\SYNERGY
        1 file(s) copied.

C:\Saxion>
```

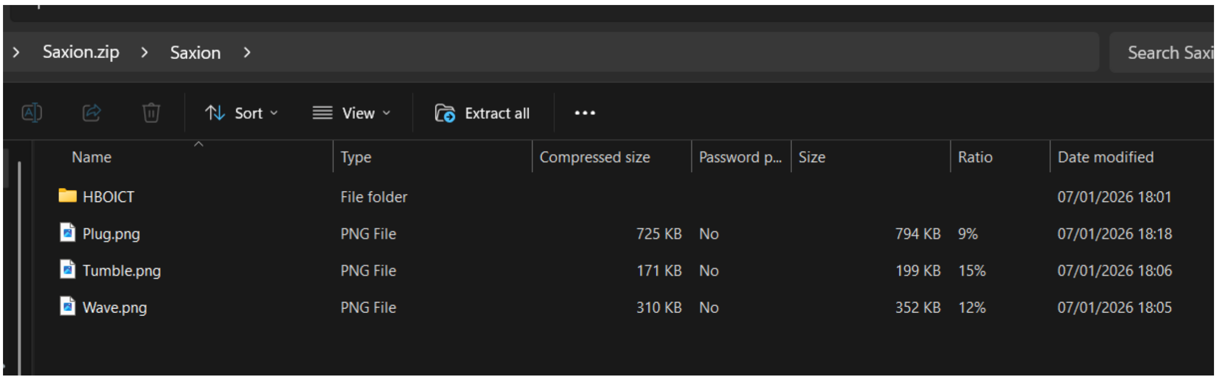
A relative path depends on the current working directory. Since the commands were executed from C:\Saxion, only the folder names were used. An absolute path would include the full path starting with the drive letter, such as C:\Saxion\IntroToProgramming.

Relevant screenshots **tree** command:



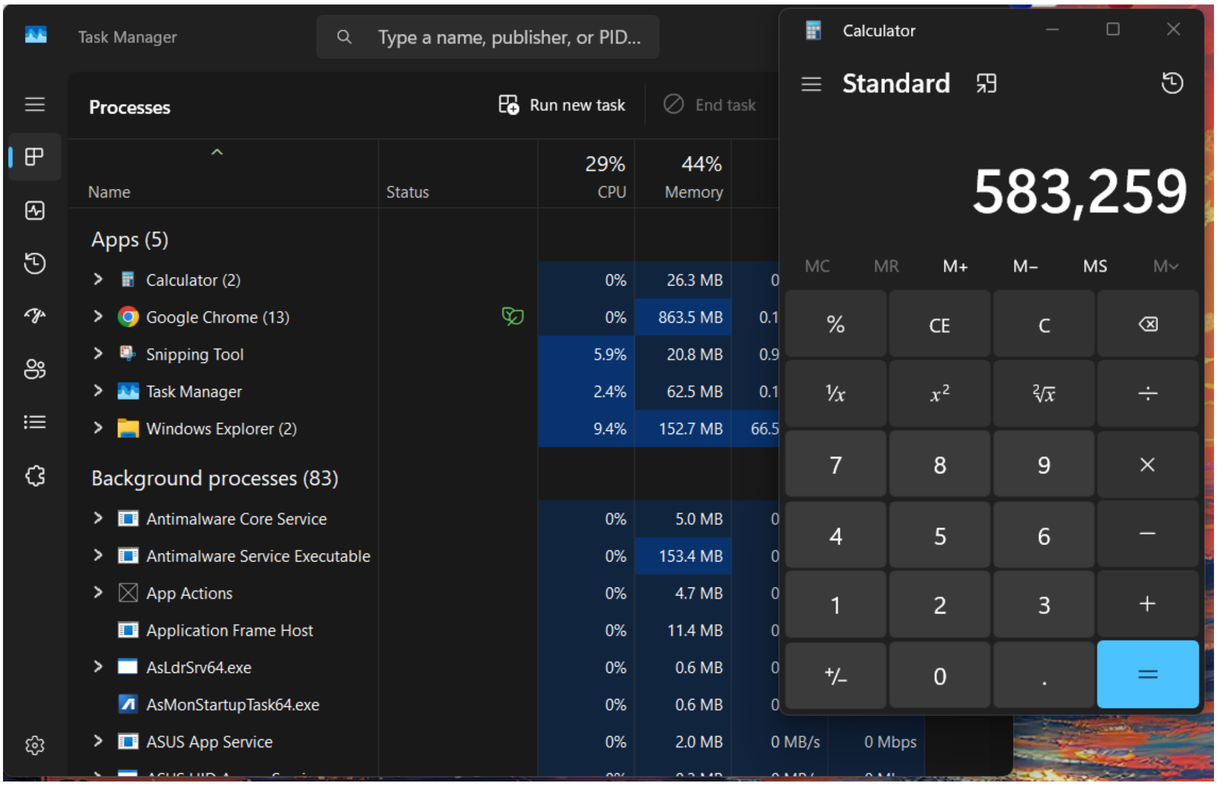


Relevant screenshots in the file explorer of the folder c:\Saxion + created zip file.



Terminating Processes

Relevant Screenshots Task Manager Window:



Install Software

Relevant screenshots that the following software is installed with winget:

Mozilla Firefox.

```
PS C:\Users\erwina> winget install -e --id Mozilla.Firefox
Found Mozilla Firefox (en-US) [Mozilla.Firefox] Version 146.0.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://download-installer.cdn.mozilla.net/pub/firefox/releases/146.0.1/win64/en-US/Firefox%20Setup%20146.0.
1.exe
  ██████████████████████████████  82.3 MB / 82.3 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Users\erwina>
```

The winget install command is used to download and install software using the Windows Package Manager. The –id option specifies the exact package identifier, which avoids ambiguity. When multiple applications have similar names. The –e option means exact, so winget will only install the package if the identifier matches exactly and will not choose a similar package.

- WinSCP

```
PS C:\Users\erwina> winget install -e --id WinSCP.WinSCP
Found WinSCP [WinSCP.WinSCP] Version 6.5.5
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://sourceforge.net/projects/winscp/files/WinSCP/6.5.5/WinSCP-6.5.5-Setup.exe/download
                              11.6 MB / 11.6 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Users\erwina> |
```

- Notepad++

```
PS C:\Users\erwina> winget install -e --id Notepad++.Notepad++
Found Notepad++ [Notepad++.Notepad++] Version 8.9
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/notepad-plus-plus/notepad-plus-plus/releases/download/v8.9/npp.8.9.Installer.x64.exe
                              6.54 MB / 6.54 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Users\erwina> winget instal -e --id
```

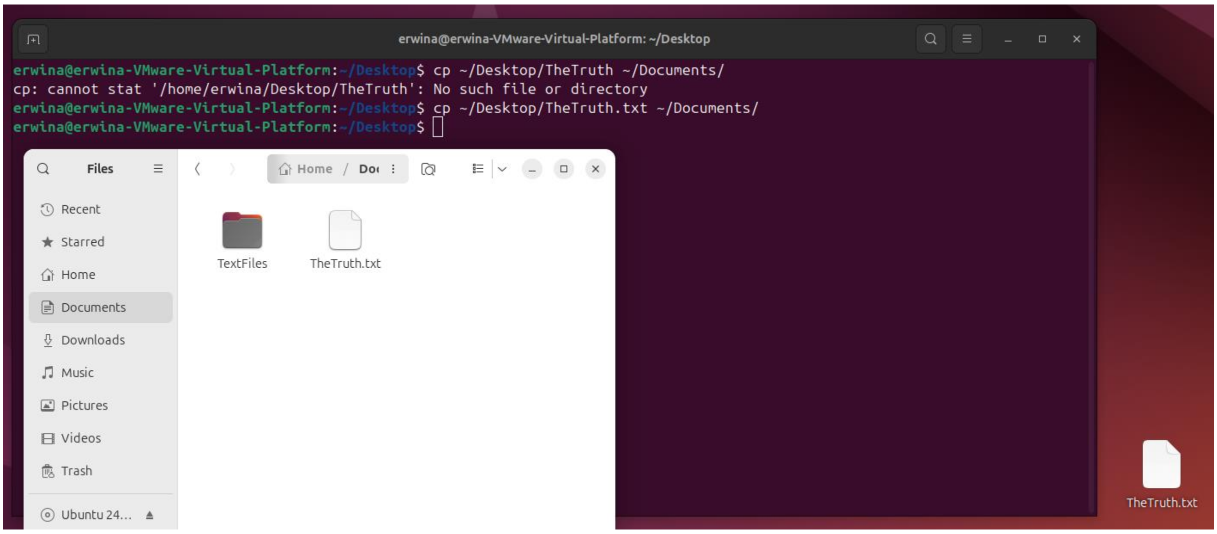
- 7zip

```
PS C:\Users\erwina> winget install -e --id 7zip.7zip
Found 7-Zip [7zip.7zip] Version 25.01
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://7-zip.org/a/7z2501-x64.exe
                              1.56 MB / 1.56 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Users\erwina>
```

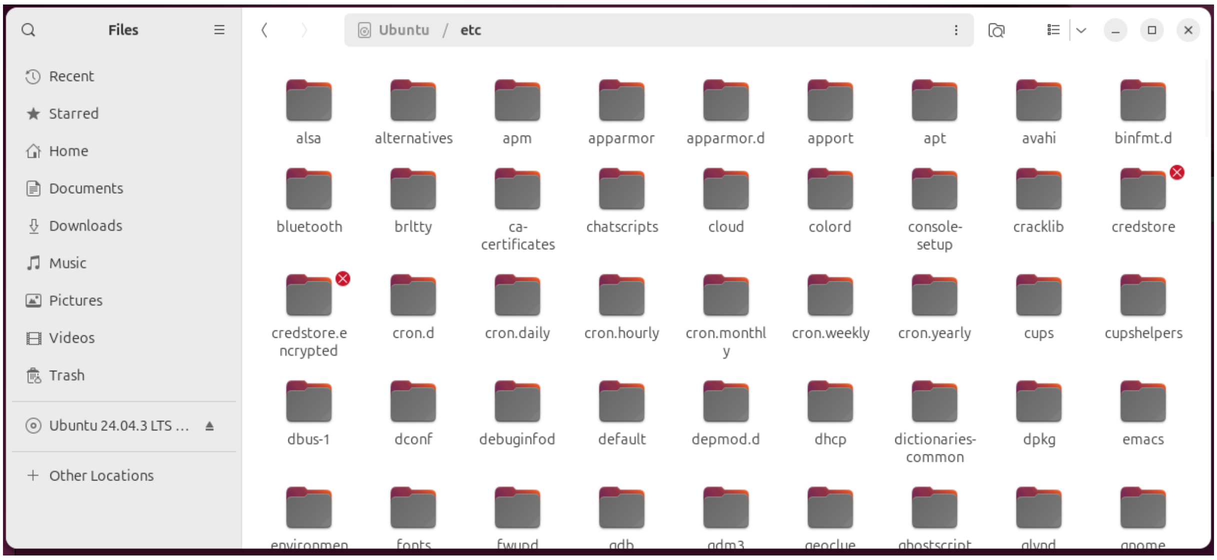
Assignment 5.4: Working with Linux

Relevant screenshots + motivation

Copying Files.



Navigating the file structure (Navigating through files)



Navigating the file structure (Terminal)

To go back to your Home folder in the terminal, you type in **cd ~.**

The difference between Linux and Windows file structure.

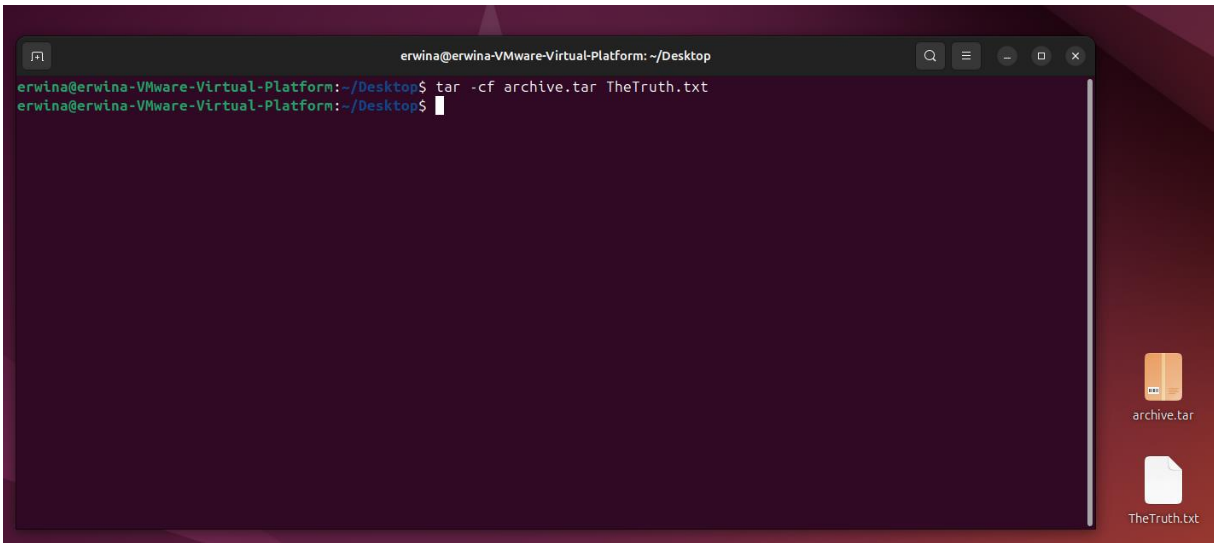
- Linux uses a single root directory (/) for all files, while Windows uses separate drive letters such as C: and D:

What is the /etc directory usually used for?

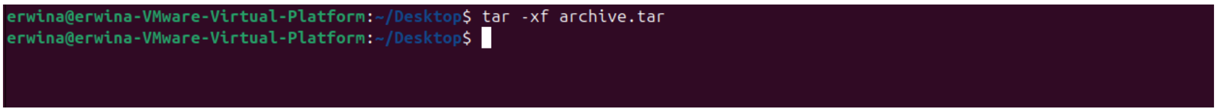
- The /etc directory is usually used to store system-wide configuration files for operating system and installed services.

Compress files

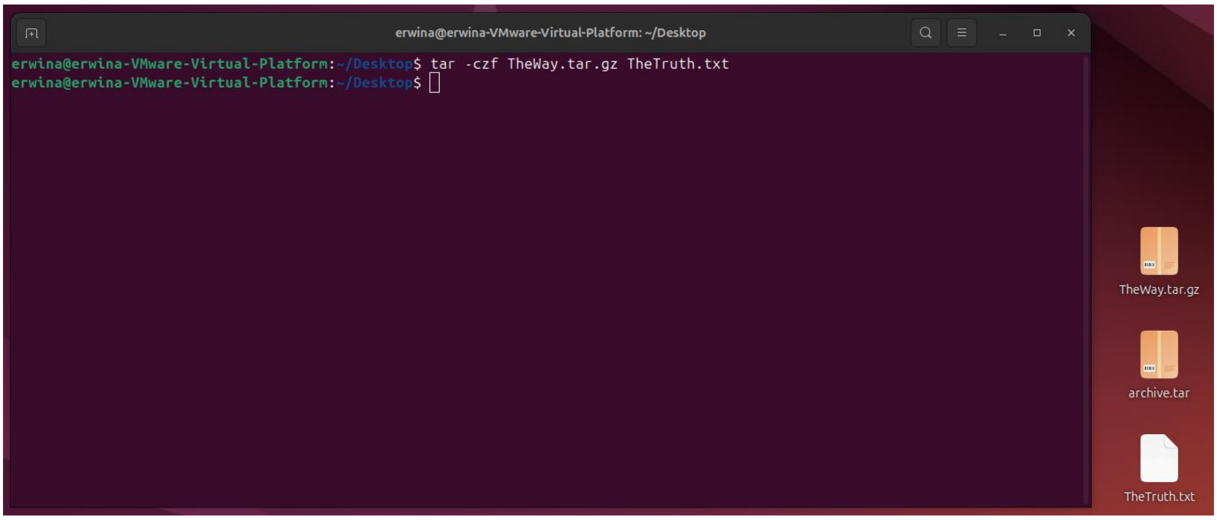
Compressing files into tar archive.



Extracting a tar file.

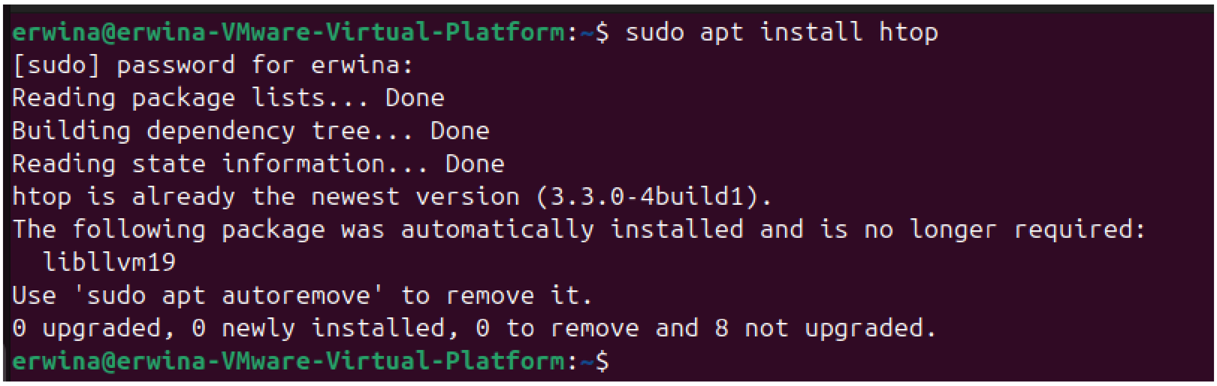


Compress a text file in a tar archive and compress it with gzip.

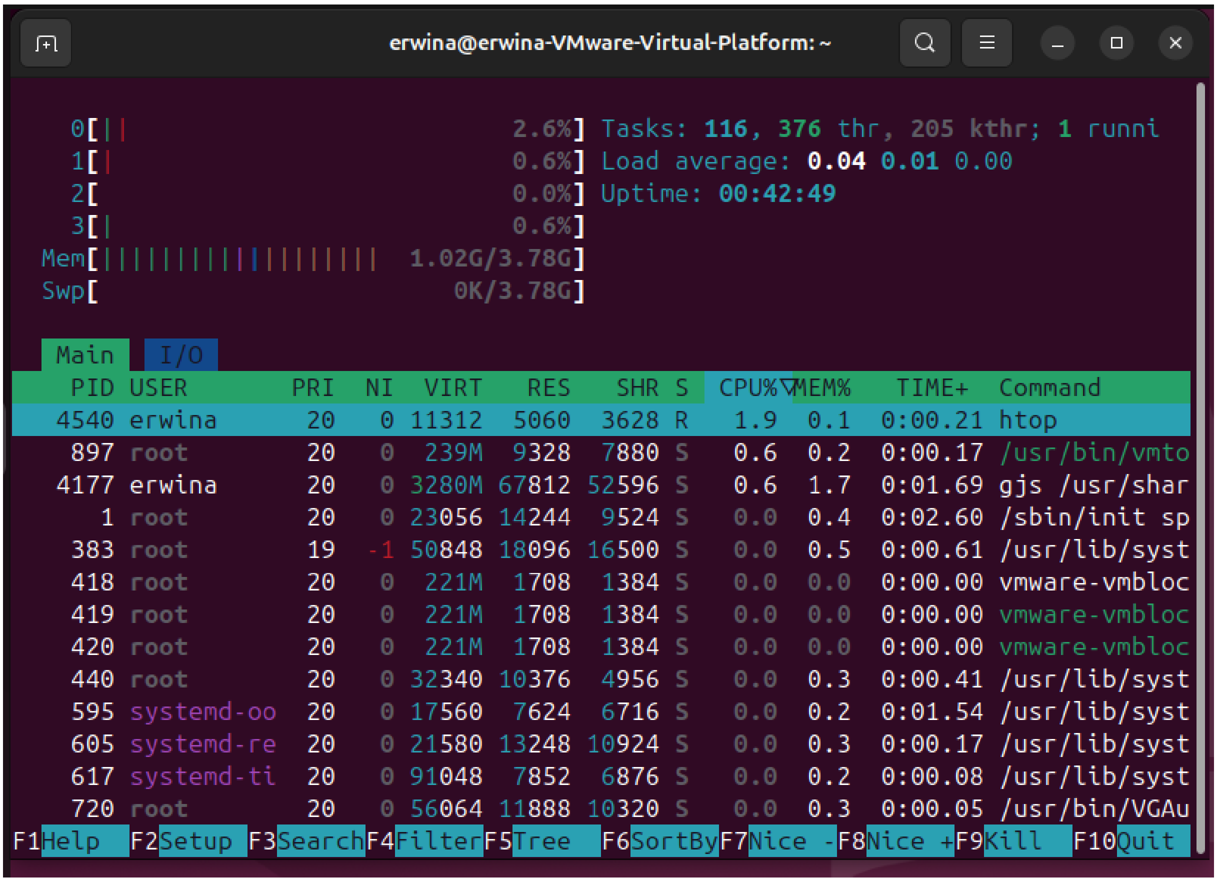


View processes

Downloaded htop.



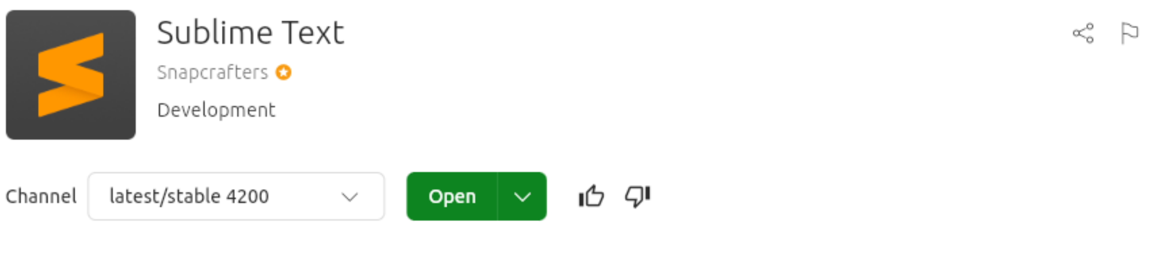
Opening htop.



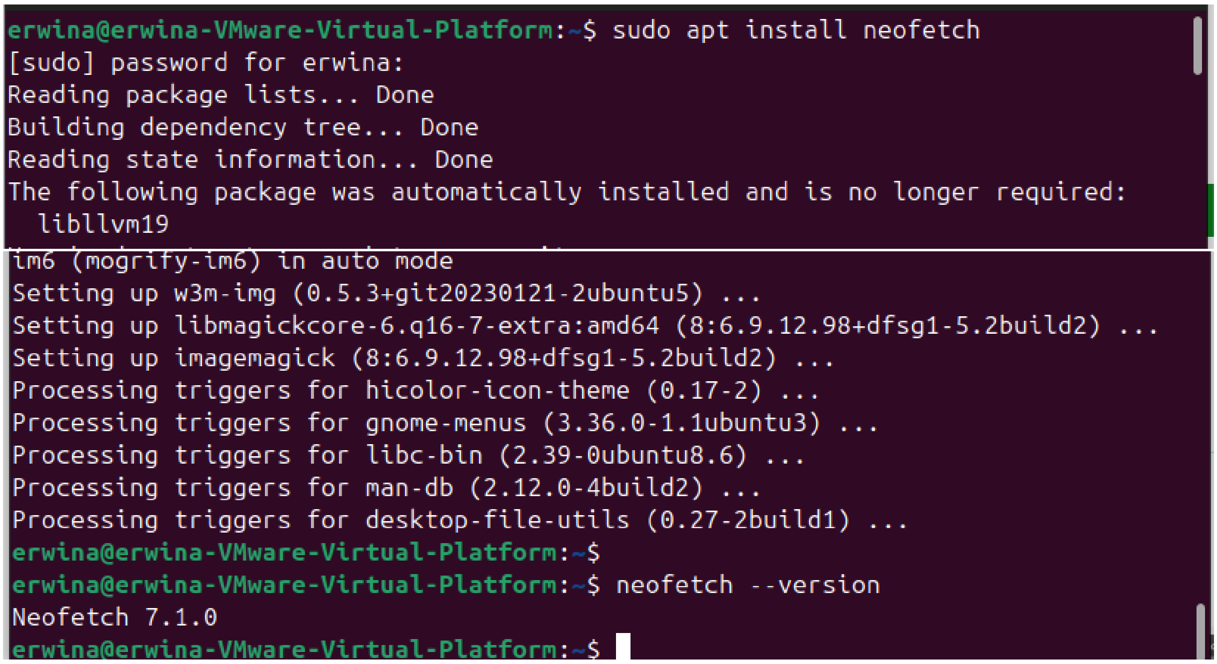
Htop shows the running process on the system in real time. It displays information such as CPU usage, memory usage, and which programs are currently active.

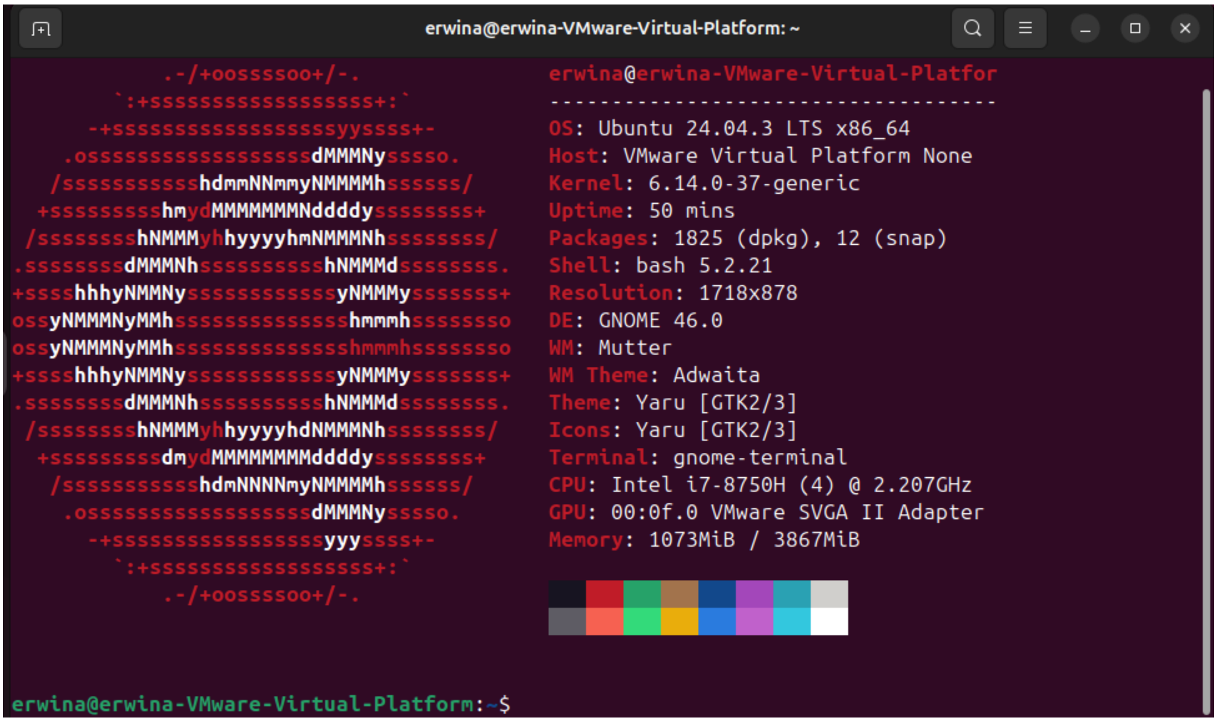
Install Software

Sublime Text



Neofetch





Neofetch displays system information such as the operating system, kernel version, hardware details, and system uptime.

Assignment 5.5: Users and permissions on Linux

Relevant screenshots + motivation

```
erwina@erwina-VMware-Virtual-Platform:~$ cd ~/Desktop
erwina@erwina-VMware-Virtual-Platform:~/Desktop$ mkdir ~/Desktop/hello
erwina@erwina-VMware-Virtual-Platform:~/Desktop$ nano ~/Desktop/hello/hello.sh
erwina@erwina-VMware-Virtual-Platform:~/Desktop$
```

```
erwina@erwina-VMware-Virtual-Platform: ~/Desktop/hello
  GNU nano 7.2                    /home/erwina/Desktop/hello/hello.sh
#!/bin/bash
echo Hello Erwina Rompas, 583259!
```

```
erwina@erwina-VMware-Virtual-Platform:~/Desktop/hello$ chmod +x hello.sh
erwina@erwina-VMware-Virtual-Platform:~/Desktop/hello$ ./hello.sh
Hello Erwina Rompas, 583259!
erwina@erwina-VMware-Virtual-Platform:~/Desktop/hello$
```

A Bash script was created in a new directory and made executable using the **chmod** command. The script was executed using ./hello.sh, which confirms that the execute permission is set correctly.

```
erwina@erwina-VMware-Virtual-Platform:~/Desktop/hello$ chmod 744 hello.sh
erwina@erwina-VMware-Virtual-Platform:~/Desktop/hello$ ./hello.sh
Hello Erwina Rompas, 583259!
```

File permissions were changed using the chmod command to control who can read and execute the script.

Assignment 5.6: View the contents of files

Relevant screenshots + motivation

```
erwina@erwina-VMware-Virtual-Platform:~/Desktop$ wc sherlock.txt
 12304 107560 607430 sherlock.txt
erwina@erwina-VMware-Virtual-Platform:~/Desktop$ wc -l sherlock.txt
12304 sherlock.txt
erwina@erwina-VMware-Virtual-Platform:~/Desktop$ wc -w sherlock.txt
107560 sherlock.txt
erwina@erwina-VMware-Virtual-Platform:~/Desktop$ wc -c sherlock.txt
607430 sherlock.txt
erwina@erwina-VMware-Virtual-Platform:~/Desktop$
```

The sherlock.txt file has 12304 lines, 107560 words, and 607430 characters in the file.

```
erwina@erwina-VMware-Virtual-Platform:~/Desktop$ grep -n "kingdom" sherlock.txt
490:"I tell you that I would give one of the provinces of my kingdom to
1124:And that was how a great scandal threatened to affect the kingdom of
erwina@erwina-VMware-Virtual-Platform:~/Desktop$
```

On the lines **490** and **1124** the word 'kingdom' appears.

```
erwina@erwina-VMware-Virtual-Platform:~/Desktop$ head -n 500 sherlock.txt | tail -n 21
"Then I shall drop you a line to let you know how we progress."

"Pray do so. I shall be all anxiety."

"Then, as to money?"

"You have _carte blanche_."

"Absolutely?"

"I tell you that I would give one of the provinces of my kingdom to
have that photograph."

"And for present expenses?"

The King took a heavy chamois leather bag from under his cloak and laid
it on the table.

"There are three hundred pounds in gold and seven hundred in notes," he
said.
```

And with the command **head –n 500 sherlock.txt | tail –n 21** you can see 10 lines above and below the word 'kingdom'. The command shows a total of 21 lines because it includes 10 lines above the given lines number, the lines containing the word 'kingdom', and 10 lines below it.

Assignment 5.7: Digital forensics

Relevant screenshots + motivation

EXIF

```
Make                                : motorola
Camera Model Name                   : moto g(6) play
```

The EXIF metadata shows that the photo was taken with a Motorola smartphone. In the Make and Camera Model Name fields you can find the brand of the phone.

```
Thumbnail Image                     : (Binary data 39433 bytes, use -b option
GPS Altitude                        : 42 m Above Sea Level
GPS Date/Time                       : 2020:11:07 14:08:57Z
GPS Latitude                        : 53 deg 11' 39.68" N
GPS Longitude                       : 6 deg 32' 12.90" E
Focal Length                        : 3.5 mm
GPS Position                        : 53 deg 11' 39.68" N, 6 deg 32' 12.90" E
Light Value                         : 7.7
```

The GPS coordinates are known. The image contains GPS metadata, including lattitude and longitude values. These coordinates indicate the location where the photo was taken. Which is in the province Groningen, near the City Haren.

Filename extensions

```
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ file oldcar
oldcar: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segme
nt length 16, Exif Standard: [TIFF image data, big-endian, direntries=10, man
ufacturer=motorola, model=moto g(6) play, xresolution=160, yresolution=168, r
esolutionunit=2, software=aljeter-user 9 PPPS29.55-35-18-7 6a0d0 release-keys
, datetime=2020:11:07 15:08:57, GPS-Data], baseline, precision 8, 4160x3120,
components 3
erwina@erwina-VMware-Virtual-Platform:~/Downloads$
```

I removed the extension of the oldcar.jpg image. Ubuntu still knows it as a JPEG/JPG image.

BASE64

```
rwina@erwina-VMware-Virtual-Platform:~/Downloads$ base64 -d email-base64.txt > decoded_output
rwina@erwina-VMware-Virtual-Platform:~/Downloads$ file decoded_output
lecoded_output: GIF image data, version 89a, 108 x 52
rwina@erwina-VMware-Virtual-Platform:~/Downloads$ mv decoded_output decoded.gif
rwina@erwina-VMware-Virtual-Platform:~/Downloads$
```

The text file email-base64.txt is decoded with the base64 –d command, and the output redirected to a new file. (decoded_output). After decoding the file command was used to identify the actual file type, which showed that the decoded file is a GIF image.

The file was then renamed using the mv command so it could be opened correctly as an image.

Assignment 5.8: Steganography

Relevant screenshots + motivation

```
erwina@erwina-VMware-Virtual-Platform:~$ cd ~/Downloads
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ chmod +x apple2.jpg
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ steghide extract -sf apple2.jpg -xf apple3.txt
Enter passphrase:
wrote extracted data to "apple3.txt".
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ cat apple3.txt
Hello class.
You have almost completed Week 5.

erwina@erwina-VMware-Virtual-Platform:~/Downloads$
```

To make the file executable, I used the chmod +x command and extracted the hidden file into a new text file.

Assignment 5.9: Capture disk images

Make relevant screenshots + motivation:

- Proof that the Debian 13 server stored a back-up image of the Ubuntu 24.04 Desktop VM.

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
erwina@erwina:~$ ssh erwina@192.168.139.131
erwina@192.168.139.131's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

60 updates can be applied immediately.
11 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

17 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Jan  8 17:31:18 2026 from 192.168.139.133
erwina@erwina-VMware-Virtual-Platform:~$
```

```
       valid_lft forever preferred_lft forever
erwina@erwina-VMware-Virtual-Platform:~$ ^C
erwina@erwina-VMware-Virtual-Platform:~$ sudo dd if=/dev/sda bs=4M status=progress | gzip | ssh erwina@192.168.139.133
"cat > /srv/images/ubuntu2404_vm.img.gz"
erwina@192.168.139.133's password:


Permission denied, please try again.
erwina@192.168.139.133's password:
37035704320 bytes (37 GB, 34 GiB) copied, 1337 s, 27.7 MB/s
```

The Debian 13 server stores the compressed disk image of the Ubuntu 24.04 in /srv/images/. The screenshot shows the image being streamed from the Debian server and successfully restored, confirming the backup exists and is valid.

- Proof that you can restore the back-up image into an empty VM.

```
ubuntu@ubuntu:~$ ssh erwina@192.168.139.133 "cat /srv/images/ubuntu2404_vm.img.gz" | gzip -d |
sudo dd of=/dev/sda bs=4M status=progress
The authenticity of host '192.168.139.133 (192.168.139.133)' can't be established.
ED25519 key fingerprint is SHA256:IwxT321d590qyNcMPuSWBsLNRCVMjgV0KH7mbAkjdBY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.139.133' (ED25519) to the list of known hosts.
erwina@192.168.139.133's password:
21394849792 bytes (21 GB, 20 GiB) copied, 342 s, 62.6 MB/s
dd: error writing '/dev/sda': No space left on device
0+647682 records in
0+647681 records out
21474836480 bytes (21 GB, 20 GiB) copied, 343.599 s, 62.5 MB/s
ubuntu@ubuntu:~$
```

This command restores a compressed disk image directly onto the VM's primary disk (/dev/sda), overwriting it completely. After the restore, the VM booted successfully, confirming the backup was correctly restored.